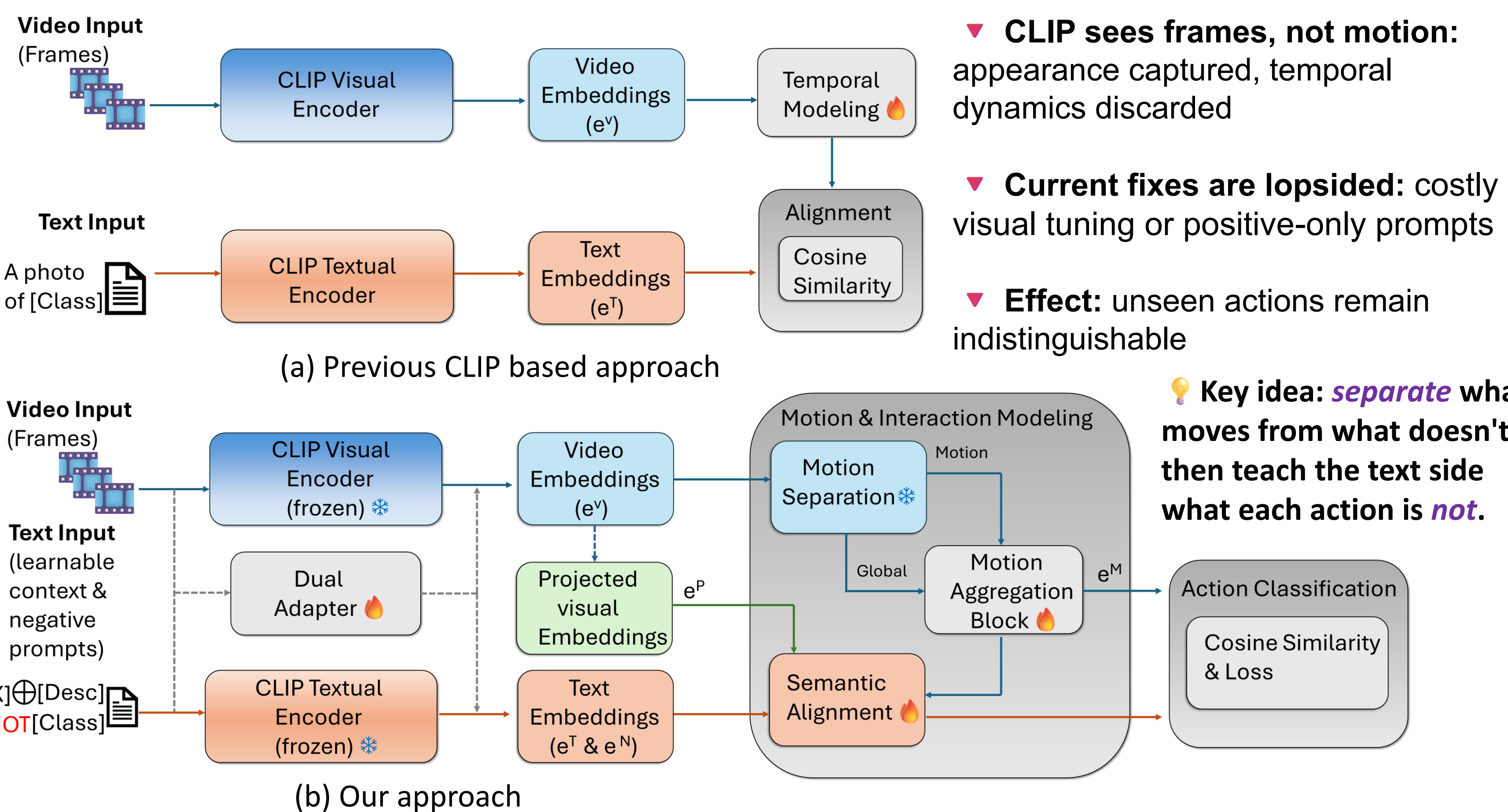


MOTION-GUIDED SEMANTIC ALIGNMENT WITH NEGATIVE PROMPTS FOR ZERO-SHOT VIDEO ACTION RECOGNITION

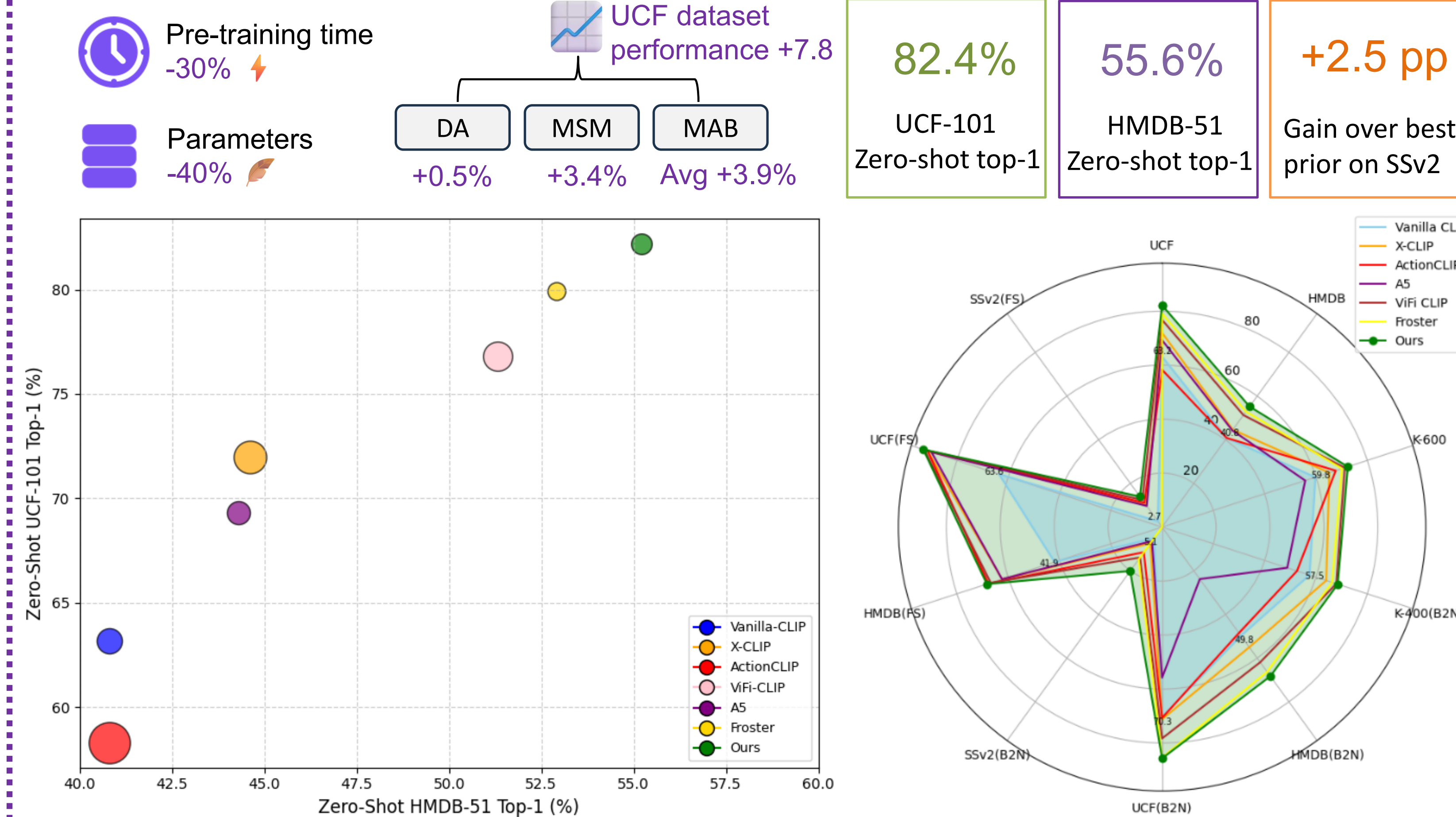
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Background & Motivations



Performance



Quantitative Results

Generalizing to Unseen Domains

Method	Publication	HMDB-51	UCF-101	K-600
Methods with Vision Training				
ER-ZSAR [3]	ICCV'21	35.3 ± 4.6	51.8 ± 2.9	42.1 ± 1.4
JigSawNet [4]	TIP'19	39.3 ± 3.9	56.8 ± 2.8	—
Methods with Vision-Language Training				
A5 [5]	ECCV'22	44.3 ± 2.2	69.3 ± 4.2	—
X-CLIP [20]	ECCV'22	46.3 ± 0.6	70.3 ± 2.3	67.1 ± 1.0
Vita-CLIP [7]	CVPR'23	48.6 ± 0.6	75.0 ± 0.6	—
SDR-CLIP [13]	ACC'24	52.7 ± 1.4	75.3 ± 3.2	—
GIL [16]	ACC'24	53.9 ± 1.4	79.4 ± 1.4	—
TP-CLIP [9]	CVPR'25	54.1 ± 1.2	81.1 ± 1.2	—
CLAVAR [10]	arXiv'25	54.1 ± 2.4	78.6 ± 1.7	—
STDD [14]	AAAI'25	55.9 ± 0.2	85.2 ± 1.2	75.1 ± 0.7
ZAR [11]	EI'25	54.2 ± 0.8	77.4 ± 0.8	70.5 ± 0.4
Ours	—	55.2 ± 1.1	82.2 ± 0.5	72.4 ± 0.5

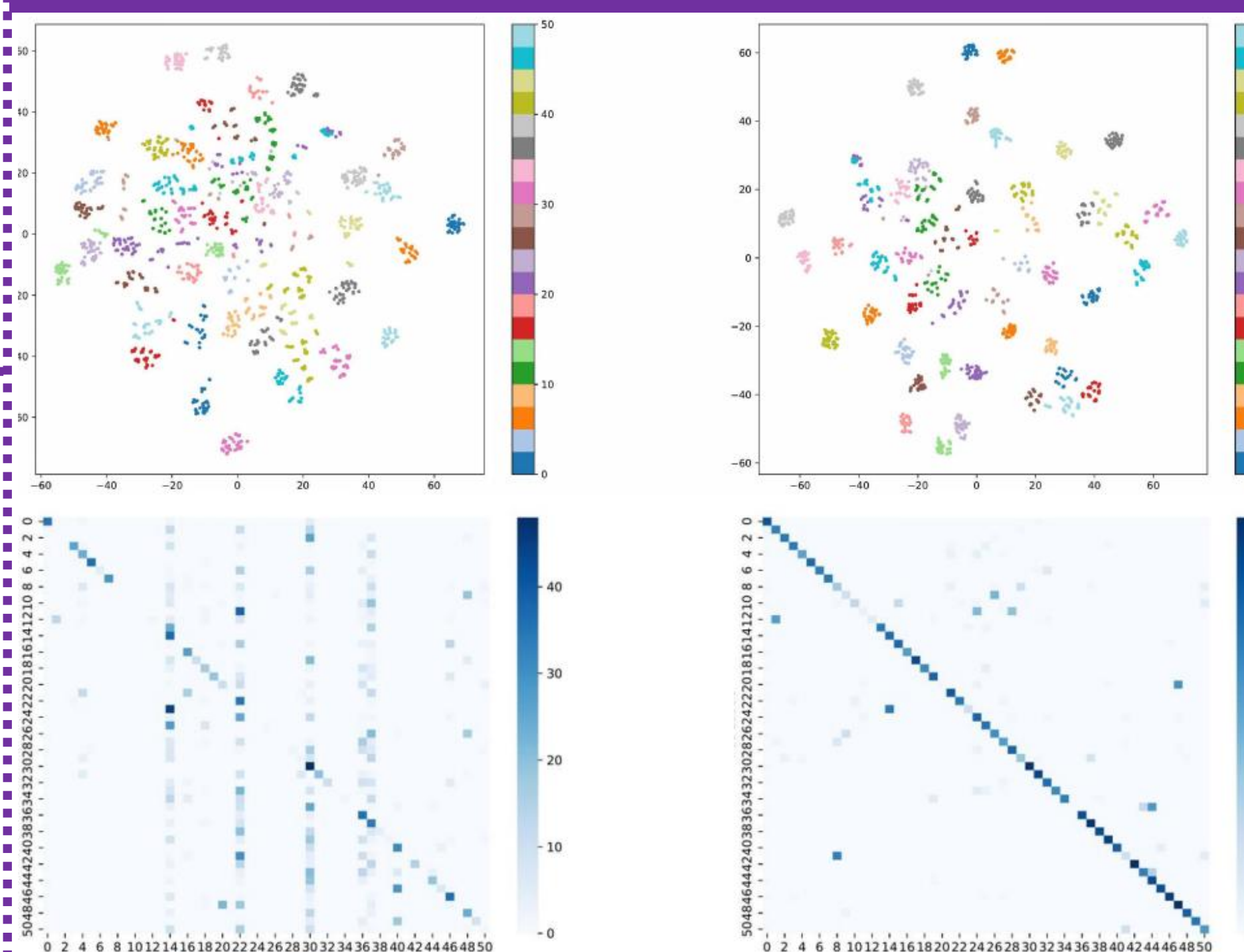
- On HMDB-51, ours **tops all methods** without vision pre-training — closing the gap with costly vision-tuned approaches.
- On Kinetics-600, **negative prompts prove essential** at scale — fine-grained classes that prior methods confuse are now separable.

Ablation Study

DA	Component / Setting			Zero-shot (%)		
	MSM	MAB	Splitting	HMDB-51	UCF-101	SSv2
✓	Baseline (Only CLIP)			36.3	58.3	5.1
			Offsets	(+1.1%)	(+0.5%)	(+41.2%)
	✓		Offsets	(-0.8%)	(+3.4%)	(+27.5%)
	✓	✓	Offsets	(+11.0%)	(+6.3%)	(+56.9%)
	✓	✓	Offsets	(+23.1%)	(+13.4%)	(+62.7%)
	✓	✓	Fixed	(+13.3%)	(+1.2%)	(+15.3%)

All three components contribute, but MAB + splitting delivers the largest leap — especially on SSv2, where actions share **identical objects** and only differ in **motion direction**, confirming that motion aggregation is the key bottleneck.

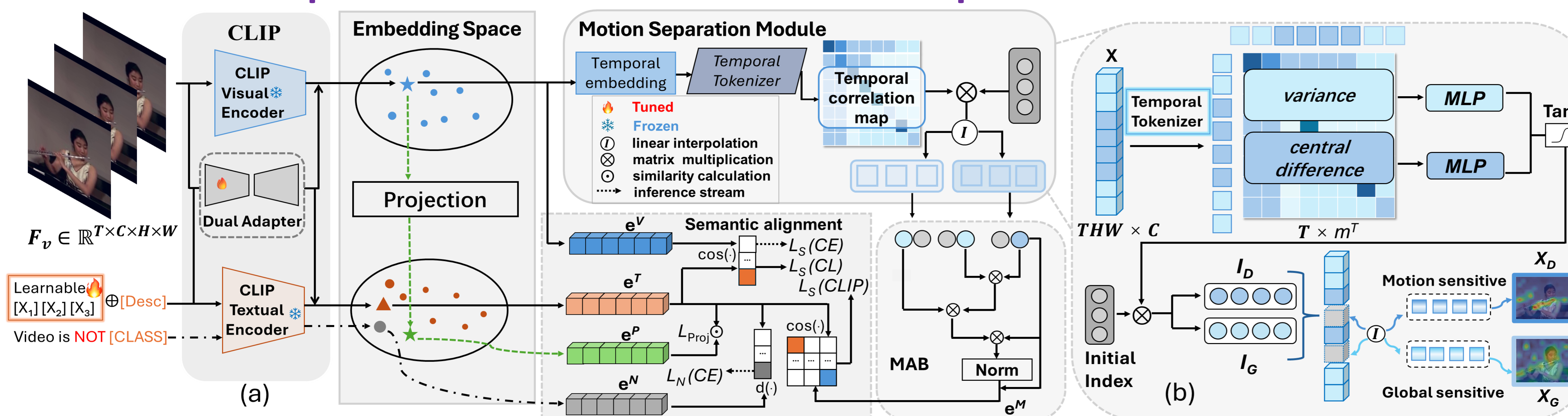
Visualization



Contributions

- Explicit motion-static disentanglement makes CLIP more suitable for zero-shot video understanding
- Negative prompts + semantic alignment improve discrimination and generalization to unseen actions

Pipeline Of CLIP-based Video-Text Representation

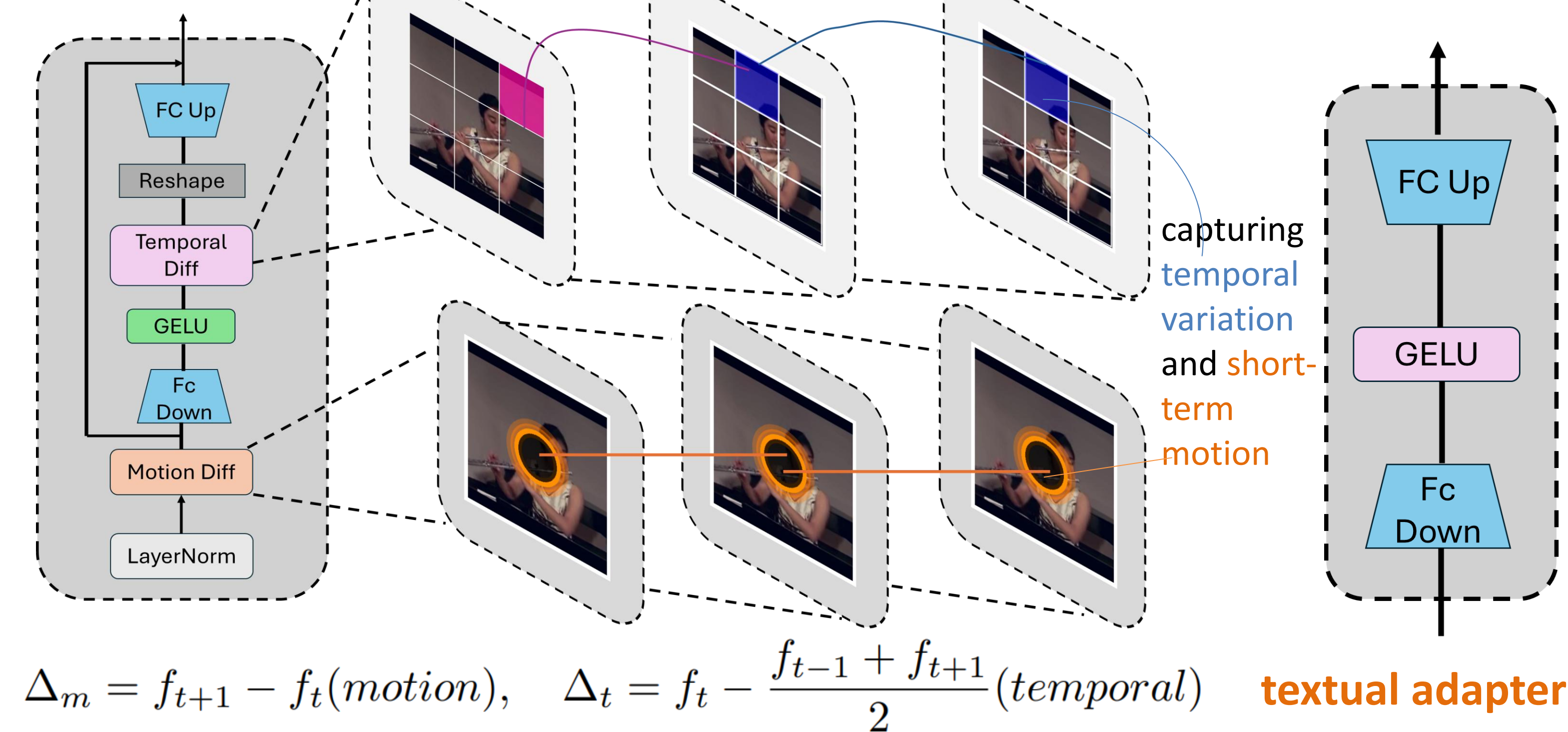


Motion Separation Module (MSM): disentangles motion-sensitive from global-static features adaptively.

Motion Aggregation Block (MAB): fuses motion and context via gated cross-attention without re-coupling redundant noise

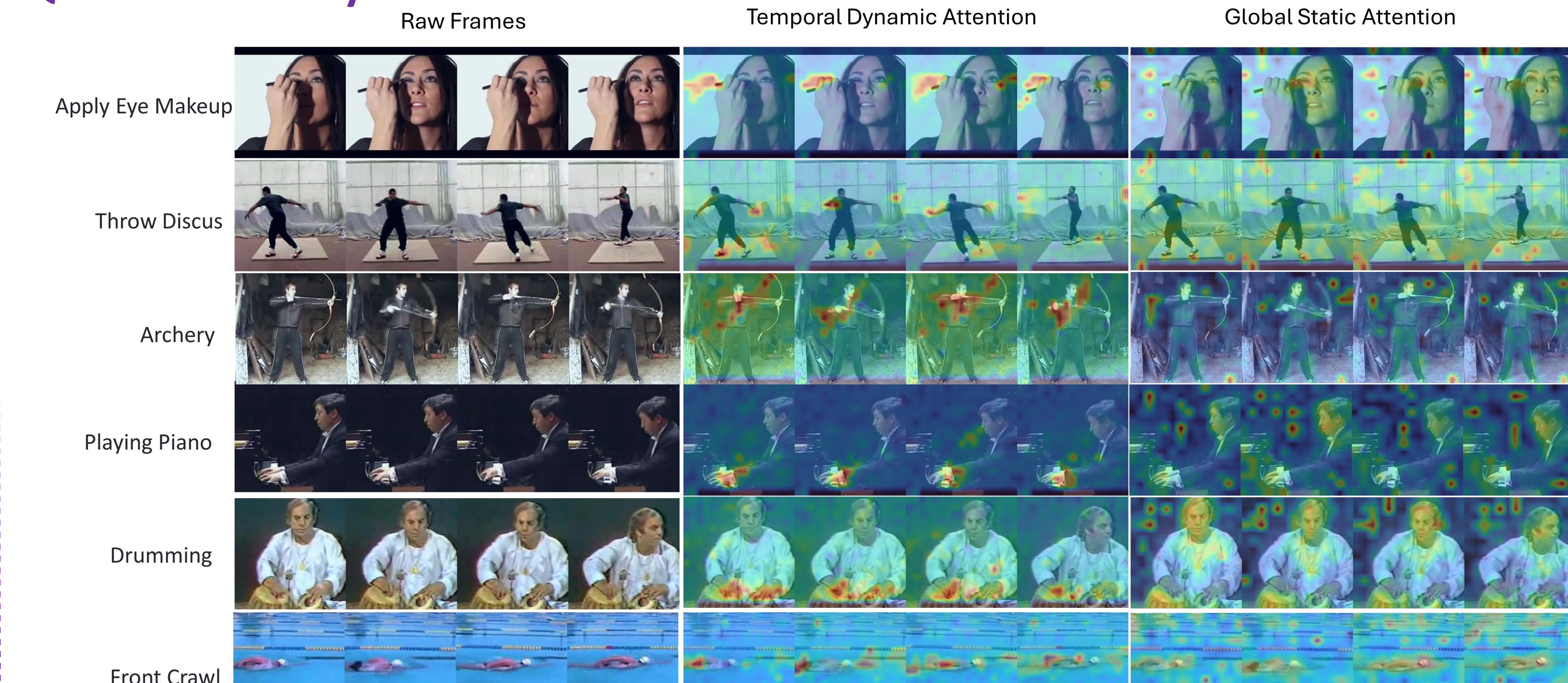
Class-dependent negative prompting: enlarges unseen-class margins by modelling "what a class is not."

visual adapter



The Dual Adapters

Qualitative Analysis Of UCF dataset



$$\Delta_m = f_{t+1} - f_t(\text{motion}), \quad \Delta_t = f_t - \frac{f_{t-1} + f_{t+1}}{2}(\text{temporal})$$

textual adapter